

Sparse and Deep Gaussian Process Closure Modeling for Non-Stationary Two-Dimensional Vorticity Flows Past Idealized Obstacles

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Abstract—Large-eddy simulation (LES) reduces the cost of turbulent-flow prediction by resolving energetic motions while modeling the effects of unresolved scales through a subgrid-scale (SGS) closure. Its accuracy therefore depends on how well the closure represents fine-scale effects on the resolved flow, a particular challenge in nonstationary obstacle flows where separation and wake dynamics are time-dependent and the filtering operation interacts with immersed boundaries. To address this challenge, we develop a generalized data-driven SGS closure for two-dimensional incompressible flow past obstacles with dynamic inflow velocity fields. The closure forcing is modeled with a compact convolutional neural network (CNN), trained on filtered direct numerical simulation (fDNS) fields and conditioned on the instantaneous Reynolds number and its recent time history, while a sparse variational Gaussian process (SVGP) head models the CNN residual and provides uncertainty estimates. Training spans stationary and nonstationary flows past bluff and streamlined bodies, with generalization evaluated across unseen geometries, Reynolds-number regimes, and forcing histories. We find that *a priori* predictions remain strongly correlated with the reference closure forcing, while *a posteriori* LES rollouts remain stable and recover resolved dynamics, integral quantities, and turbulent statistics across unseen flow configurations. After the LES and fDNS trajectories decorrelate, the learned closure continues to reproduce the statistical behavior of the fDNS reference.

Index Terms—large-eddy simulation, subgrid-scale closure, machine learning, Gaussian process, nonstationary obstacle flows

I. INTRODUCTION

Turbulent flows past obstacles involve boundary layers, separated shear layers, and wakes spanning a wide range of spatial and temporal scales [1]. Direct numerical simulation (DNS) resolves these dynamics without turbulence modeling but is often prohibitively expensive at the Reynolds numbers (Re) and ensemble sizes of interest. Reynolds-averaged approaches reduce this cost by solving for the mean flow and modeling the effects of turbulent fluctuations through Reynolds stresses. However, this averaging can limit their ability to represent strongly unsteady separated flows [2]. Large-eddy simulation [3, 4] provides a good compromise in which energetic motions are resolved and fine-scale effects are represented through a subgrid-scale closure (SGS). Its predictive accuracy therefore depends critically on the closure model.

Many established SGS closures use an eddy-viscosity formulation. The static Smagorinsky model prescribes an eddy viscosity coefficient C_s , while the dynamic variant estimates it from the resolved flow through the Germano identity [5, 6]. However, eddy-viscosity closures can exhibit weak point-wise correlations with reference SGS stresses and, when strictly dissipative, cannot represent local backscatter [7, 8]. Additionally,

dynamic formulations usually require averaging or clipping for stable *a posteriori* LES [9].

Data-driven methods offer an attractive alternative to fixed analytical closure forms [10]. Machine-learned SGS models have been studied in homogeneous isotropic turbulence [11], two-dimensional (2D) turbulence [12], turbulent channel flow [13], and quasi-geostrophic and ocean flows [14], demonstrating that unresolved fine-scale contributions can be accurately learned from fDNS data and, in several cases, successfully deployed *a posteriori*.

Even though applications to flows past solid obstacles have received comparatively less attention, some promising neural SGS models have been developed for a circular cylinder [15], a backward-facing step [16], and rectangular bluff bodies [17], while graph-based approaches have been proposed to improve transfer across geometries [18]. Several of these studies also provide evidence of generalization beyond the training configurations. For example, Kim et al. [15] train on fDNS fields at $Re_D = 3900$ and obtain accurate *a posteriori* statistics at $Re_D = 5000$ and 10,000. Sirignano and MacArt [17] similarly evaluate their model at unseen Re and bluff-body geometries. However, these obstacle-flow SGS models have primarily considered stationary conditions.

In this study, we develop generalizable, data-driven SGS closure models for 2D incompressible flows with dynamic forcing. We focus on flows past obstacles with variable background velocity fields and, consequently, time-dependent inflow Re. The forcing ensemble includes periodic modulation, acceleration, and stochastic variability, representing nonstationary phenomena typical of accelerated bluff-body flows [19], tidally modulated currents past topography [20], and stochastic models of turbulent velocity fluctuations [21]. The closure must therefore remain accurate as the resolved flow responds to both the instantaneous forcing and its time history. To achieve this, our closure forcing is modeled by a compact CNN conditioned on the instantaneous Re and its recent history, with an SVGP head [22] modeling the CNN residual and uncertainty. The closure is evaluated both *a priori* and *a posteriori* under distribution shifts, with emphasis on generalization, long-horizon stability, trajectory correlation, and recovery of integral and statistical quantities before and after the LES and fDNS trajectories decorrelate [23, 24].

Flows near immersed boundaries can introduce challenges for spatial SGS filtering: the filter can act on both the flow and boundary treatment. To address this, Kim et al. [15] use a one-sided box filter near the cylinder and exclude solid points from training, while Sirignano and MacArt [17] instead retain the boundary and filter-commutation contributions and account

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for their combined effect by back-propagating through the LES solver. Here, a Brinkman volume penalization represents the obstacle [25], and a translation-invariant Gaussian convolution filters the DNS fields. The contributions of advective SGS and filter-obstacle interactions can therefore be evaluated separately and point-wise from DNS and fDNS data, yielding an explicit supervised closure target.

In what follows, Secs. II and III present the governing equations, numerical scheme, filtering procedure, and the closure forcing. Sec. IV describes the nonstationary training (IND) and out-of-distribution (OOD) evaluation ensembles. Sec. V details the CNN-SVGP closure and its training. Sec. VI presents the a priori and a posteriori results.

II. GOVERNING EQUATIONS AND NUMERICAL SCHEME

A. Governing Equations

We consider 2D incompressible flows governed by the Navier-Stokes equations in the vorticity-stream-function form, and forced with a dynamic streamwise background velocity $U(t)$. Defining the vorticity ω , stream-function ψ , and total velocity $\mathbf{u} = (u, v)$:

$$\begin{aligned} \frac{\partial \omega}{\partial t} + J(\psi, \omega) + U(t) \frac{\partial \omega}{\partial x} &= \nu \nabla^2 \omega + F, \\ (u, v) &= (U(t) - \psi_y, \psi_x), \quad \omega = \nabla^2 \psi, \end{aligned} \quad (1)$$

where $J(\psi, \omega)$ is the nonlinear advection operator and F is the forcing defined in Sec. II-C. For an obstacle of characteristic length ℓ , the instantaneous background Re is $\text{Re}(t) = U(t)\ell/\nu$.

B. Spatial and Temporal Numerical Discretizations

The software of [25] is employed. Spatial derivatives are evaluated using a Fourier pseudo-spectral discretization in a square domain $\Omega = [0, L_\Omega] \times [0, L_\Omega]$ with $L_\Omega = 8\pi$, and nonlinear products are formed in physical space and dealiased with the 3/2 zero-padding rule [26]. Near the largest resolved wavenumbers, we apply the Hou–Li filter to smoothly damp the near-Nyquist tail: $\hat{G}_{\text{HL}}(k) = \exp[-36(|k|/k_{\text{max}})^{36}]$, where k_{max} is the maximum resolved wavenumber [27].

For time-stepping, we use a second-order Adams–Bashforth–Crank–Nicolson scheme [28, 29] where viscous diffusion is treated implicitly, while advection, Brinkman penalization, and sponge forcing are advanced explicitly as performed in [25].

C. Boundary and Obstacle Treatment

Most Fourier pseudo-spectral discretization assumes periodicity in both coordinate directions, whereas the present simulations require aperiodic streamwise inflow and outflow conditions. Following [25], these are imposed with a penalized sponge region near the streamwise boundaries,

$$\begin{aligned} F_{\text{sponge}}(\chi_{\text{sponge}}, \omega) &= -\frac{\chi_{\text{sponge}}(\mathbf{x})}{\eta_{\text{sponge}}} (\omega - \omega_{\text{sponge}}), \\ \chi_{\text{sponge}}(\mathbf{x}) &= \begin{cases} 1, & \mathbf{x} \in \partial\Omega, \\ 1 - d(x, y)/d_s, & \mathbf{x} \in \Omega_s, \\ 0, & \mathbf{x} \in \Omega \setminus \Omega_s \end{cases}, \end{aligned} \quad (2)$$

where Ω_s denotes the sponge region, $d(x, y)$ the distance from the domain boundary, d_s the sponge thickness, η_{sponge} the sponge parameter, and ω_{sponge} the target vorticity.

The no-slip condition on the immersed obstacle is enforced with a Brinkman volume penalization [30],

$$\begin{aligned} F_{\text{obs}}(\chi_{\text{obs}}, \mathbf{u}) &= -\nabla \times \left[\frac{\chi_{\text{obs}}(\mathbf{x})}{\eta} (\mathbf{u} - \mathbf{u}_{\text{obs}}) \right] \cdot \hat{\mathbf{e}}_z, \\ \chi_{\text{obs}}(\mathbf{x}) &= \begin{cases} 1, & \mathbf{x} \in \Omega_{\text{obs}}, \\ 0, & \mathbf{x} \notin \Omega_{\text{obs}}, \end{cases} \quad \eta = C_\eta \Delta t, \end{aligned} \quad (3)$$

where Ω_{obs} denotes the solid region and C_η the porosity coefficient. Since our obstacles are stationary, $\mathbf{u}_{\text{obs}} = 0$, and the total forcing in Eq. (1) is $F = F_{\text{obs}} + F_{\text{sponge}}$.

The force exerted on the obstacle is obtained from the penalization integral [31],

$$\mathbf{F}_{\text{body}} = \int_{\Omega} \frac{\chi_{\text{obs}}}{\eta} (\mathbf{u} - \mathbf{u}_{\text{obs}}) dA, \quad (4)$$

with drag and lift given by its streamwise and cross-stream components.

III. FILTERING AND CLOSURE FORCING

A. Gaussian Filtering

The filtered DNS (fDNS) fields are obtained by applying a Gaussian filter [32] to the DNS fields. The DNS and LES grid spacings are

$$\begin{aligned} \Delta_{\text{DNS}} &= \frac{L_\Omega}{N_{\text{DNS}}}, \quad N_{\text{DNS}} = 2048, \\ \Delta_{\text{LES}} &= \frac{L_\Omega}{N_{\text{LES}}}, \quad N_{\text{LES}} = 256. \end{aligned} \quad (5)$$

The two Gaussian smoothing operations applied in Fourier space combine into

$$\begin{aligned} \hat{q}(\mathbf{k}) &= \exp\left(-\frac{k^2 \sigma_G^2}{6}\right) \hat{q}(\mathbf{k}), \quad \sigma_G^2 = \sigma_1^2 + \sigma_2^2, \\ \sigma_1 &= 1.5 \Delta_{\text{DNS}}, \quad \sigma_2 = \delta_{\text{LES}} \Delta_{\text{LES}}. \end{aligned} \quad (6)$$

Equivalently, $\hat{G} = \exp(-k^2 \bar{\Delta}^2/24)$ with $\bar{\Delta} = 2\sigma_G$. The filter is translation invariant and therefore commutes with spatial differentiation. We use $\delta_{\text{LES}} = 1.5$, corresponding to $\bar{\Delta}/\Delta_{\text{LES}} = 3.02$.

B. Filtered Equations and Reference SGS Closure

Applying the Gaussian filter in Eq. (6) to the governing equations yields

$$\begin{aligned} \frac{\partial \bar{\omega}}{\partial t} + J(\bar{\psi}, \bar{\omega}) + U(t) \frac{\partial \bar{\omega}}{\partial x} &= \nu \nabla^2 \bar{\omega} + F_{\text{obs}}(\bar{\chi}_{\text{obs}}, \bar{\mathbf{u}}) \\ &\quad + F_{\text{sponge}}(\bar{\chi}_{\text{sponge}}, \bar{\omega}) + \Pi. \end{aligned} \quad (7)$$

The reference closure term is defined as:

$$\begin{aligned} \Pi &= \underbrace{J(\bar{\psi}, \bar{\omega}) - J(\psi, \omega)}_{\Pi_{\text{SGS}}} \\ &\quad + \underbrace{F_{\text{obs}}(\chi_{\text{obs}}, \mathbf{u}) - F_{\text{obs}}(\bar{\chi}_{\text{obs}}, \bar{\mathbf{u}})}_{\Pi_{\text{B}}} \\ &\quad + \underbrace{F_{\text{sponge}}(\chi_{\text{sponge}}, \omega) - F_{\text{sponge}}(\bar{\chi}_{\text{sponge}}, \bar{\omega})}_{\Pi_{\text{sp}}}. \end{aligned} \quad (8)$$

configuration	reference length	D_χ	ν	$\text{Re}(\bar{U})$	C_η	Δt_{DNS}	Δt_{LES}
circular island	$D = 1.26$	1.26	6.44×10^{-4}	3900	1.25	2.5×10^{-4}	1.0×10^{-3}
square prism, OOD	$D = 1.26$	1.42	6.44×10^{-4}	3900	1.025	1.25×10^{-4}	1.0×10^{-3}
coastal cape	$h_c = 4.03$	2.65	6.44×10^{-4}	12500	1.025	1.25×10^{-4}	1.0×10^{-3}
coastal cape, aspect-ratio OOD	$h_c = 2.75$	2.66	6.44×10^{-4}	8500	1.025	1.25×10^{-4}	1.0×10^{-3}
DARPA SUBOFF submarine	$L = 10.60$	3.92	6.44×10^{-4}	32900	1.025	1.25×10^{-4}	1.0×10^{-3}
submarine, Re OOD	$L = 10.60$	3.92	4.19×10^{-4}	50600	1.025	1.25×10^{-4}	1.0×10^{-3}

Table I. Reference dimensions and numerical parameters for the IND and OOD simulated trajectories.

The three terms respectively represent contributions from nonlinear advection, Brinkman penalization, and sponge forcing. Since Π_{sp} is small relative to the first two, we retain

$$\Pi \simeq \Pi_{\text{SGS}} + \Pi_{\text{B}}, \quad (9)$$

with both contributions evaluated point-wise from the stored DNS and fDNS fields.

Filtering the binary obstacle mask smooths the sharp DNS obstacle interface over a narrow band, with $\bar{\chi}_{\text{obs}} = 1/2$ at the original boundary where the signed distance function $\text{sdf} = 0$, up to curvature corrections of order $O(\sigma_G \kappa)$, where κ is the curvature of the local interface. The LES Brinkman forcing is evaluated using this filtered mask directly, so the obstacle boundary is represented by the same graded transition that results from filtering the DNS equations. This preserves consistency between the filtered fields, the Brinkman forcing, and the reference closure in Eqs. (7)-(8).

IV. NONSTATIONARY TRAINING AND OOD DATA

A. Flow Configurations

The training ensemble contains three obstacle geometries: a circular island, a coastal cape, and the standard DARPA SUBOFF submarine with fairwater [33]. For comparisons and normalization across geometries, we additionally define the area-equivalent diameter

$$D_\chi = 2\sqrt{\frac{A_\chi}{\pi}}, \quad (10)$$

where A_χ is the obstacle area defined by χ_{obs} (3). The characteristic dimensions in $\text{Re}(t) = U(t)\ell/\nu$ are shown in Fig. 1. The training in-distribution and out-of-distribution configurations, together with their simulation parameters, are summarized in Table I. The DNS-to-fDNS filtering and retained training pixels are illustrated in Fig. 2.

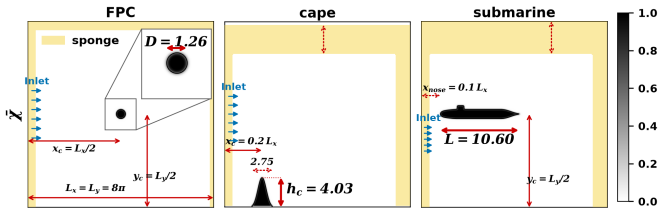


Fig. 1. Training configurations and characteristic lengths used in the definition of $\text{Re}(t)$ for the three obstacles considered.

B. Time-Dependent Inlet Forcing

The nonstationary trajectories are generated by prescribing the inflow background velocity as a function of time. The modulation laws are centered on the nominal velocity $\bar{U} = 2$

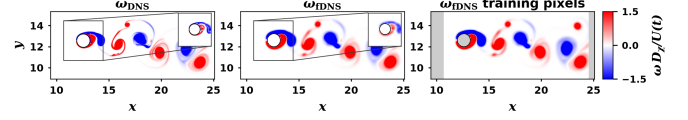


Fig. 2. DNS and fDNS vorticity fields and retained training pixels for the flow past a circular island.

with amplitude $a_U = 0.44$, such that $U(t) \in [(1 - a_U)\bar{U}, (1 + a_U)\bar{U}]$. Each simulation is first advanced with the constant inlet velocity $U(t) = \bar{U}$ for a common initialization interval $T_w = 30$, after which the prescribed modulation is applied. The forcing time scales are expressed relative to the characteristic shedding period $T_{\text{shed}} = D_\chi/[St \langle U \rangle_t]$, with Strouhal number $St = 0.21$, where $\langle U \rangle_t$ denotes the time-averaged inlet velocity over the post-initialization interval.

C. Training IND and OOD Data

The training IND database contains eleven trajectories: the circular island and coastal cape are each simulated with the five inlet functions in Fig. 3, while the submarine contributes one constant-inlet trajectory at $\text{Re}_L(\bar{U}) = 32900$. The OOD evaluation protocol is summarized in Fig. 4.

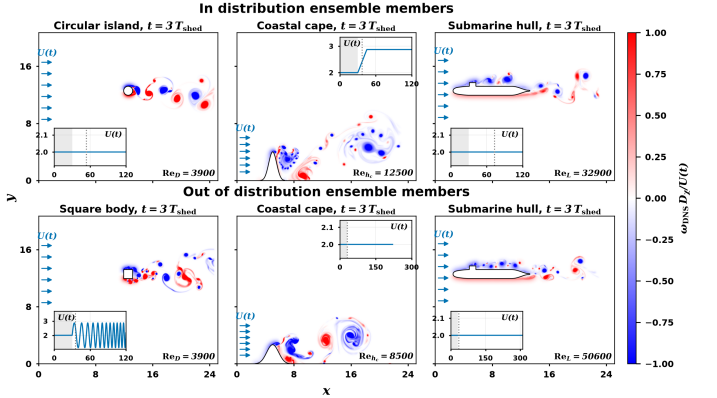


Fig. 3. IND and OOD flow configurations: body geometry, inlet forcing, and Reynolds number

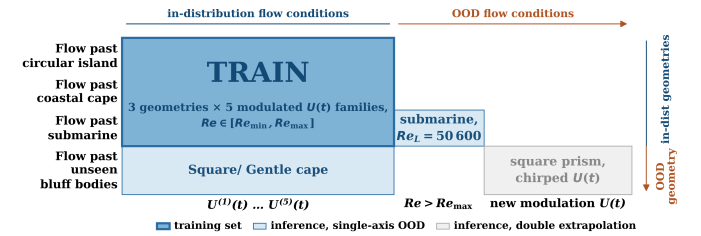


Fig. 4. OOD evaluation across geometry, Reynolds number, and time-dependent inlet forcing.

Within each in-distribution trajectory, the first 80% of the post-initialization sequence is used for training and the

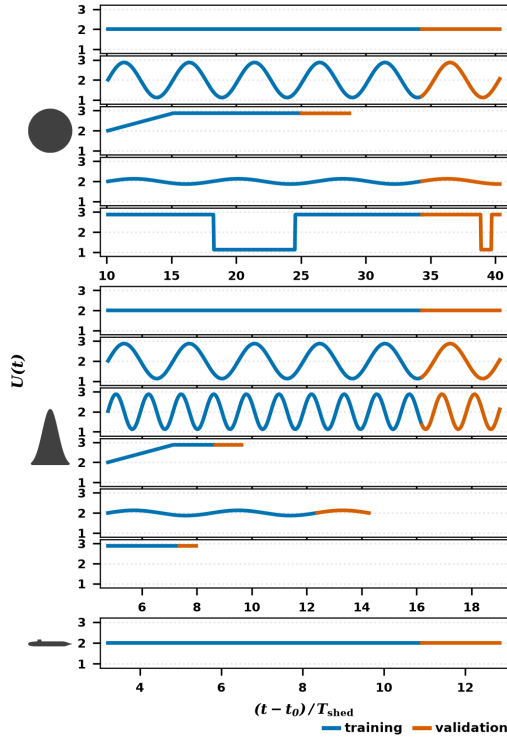


Fig. 5. Temporal split between training and validation data.

remaining 20% for validation, preventing neighboring correlated snapshots from appearing in both sets. Each trajectory contributes up to 160 training and 40 validation fields. The split is shown in Fig. 5.

V. NEURAL CLOSURE MODEL AND TRAINING

A. Inputs and Normalization

The four filtered flow channels are non-dimensionalized with the instantaneous inlet velocity and the area-equivalent diameter from Eq. (10),

$$\begin{aligned} \bar{\omega}^* &= \frac{\bar{\omega} D_\chi}{U(t)}, & \bar{u}^* &= \frac{\bar{u}}{U(t)}, \\ \bar{v}^* &= \frac{\bar{v}}{U(t)}, & \bar{\psi}^* &= \frac{\bar{\psi}}{U(t) D_\chi}. \end{aligned} \quad (11)$$

Geometry is supplied on the same Cartesian grid through a clipped signed-distance field,

$$\text{sdf}(\mathbf{x}) = \frac{\text{clip}(\text{sdf}(\mathbf{x}), -cD_\chi, cD_\chi)}{cD_\chi}, \quad c = 2, \quad (12)$$

measured from the $\bar{\chi}_{\text{obs}} = 0.5$ contour and negative inside the obstacle. Within the unclipped band, the signed distance satisfies $|\nabla \text{sdf}| = 1$, so ∇sdf gives the local wall-normal direction, while $\nabla^2 \text{sdf} = \kappa$ gives the local level-set curvature. This single channel therefore encodes distance, orientation, and curvature, allowing the CNN to distinguish geometries ranging from the circular island ($\kappa = 2/D$) to square-prism corners, where curvature becomes singular. Clipping limits this geometric information to $|\text{sdf}| \leq cD_\chi$; outside that band the sdf field is constant and its derivatives vanish. This representation is commonly used in obstacle flow predictions [34, 35].

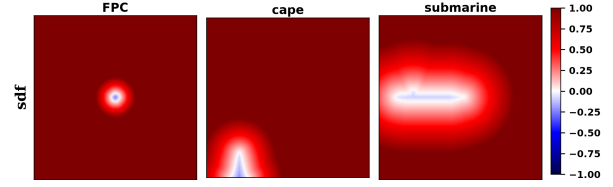


Fig. 6. Filtered obstacle mask $\bar{\chi}_{\text{obs}}$ and signed-distance field sdf used to construct the geometry inputs.

Two additional indicator planes separate the vortex-formation region, $\Omega_f : -1.5D_\chi \leq x - x_c \leq 1.5D_\chi$, and near wake, $\Omega_w : 1.5D_\chi < x - x_c \leq 6D_\chi$, where x_c denotes the obstacle reference location in the streamwise direction. Together with the four filtered flow quantities, these fields give seven spatial input channels. The filtered mask and signed-distance input are shown in Fig. 6.

The inflow history is supplied separately through

$$\zeta(t) = \frac{U(t) - \bar{U}}{a_U \bar{U}}, \quad \dot{\zeta}(t) = \frac{d\zeta}{dt}. \quad (13)$$

The derivative $\dot{\zeta}$ is averaged over one shedding-period T_{shed} and standardized with training-set statistics. After each convolutional block, $(\zeta, \dot{\zeta})$ is mapped to channel-wise scale and shift coefficients (γ, β) , which modulate the feature maps as $\mathbf{h}' = \gamma \odot \mathbf{h} + \beta$. This FiLM conditioning allows CNN features to adapt to instantaneous forcing and its recent history.

The retained target closure forcing is non-dimensionalized and standardized as

$$\Pi^* = \Pi \frac{D_\chi^2}{U(t)^2}, \quad y(\mathbf{x}, t) = \frac{\Pi^*(\mathbf{x}, t)}{\sigma_{\text{loc}}(\mathbf{x})}, \quad (14)$$

where the frozen scale σ_{loc} is a piecewise-linear profile fitted once on the training split. The domain is partitioned into three regions $g \in \{\text{formation, near wake, wake}\}$, each carrying its own normalized coordinate $\xi \in [0, 1]$, discretized into $N_b = 24$ uniform bins with centers ξ_b . With $\mathcal{B}_{r,b}$ the training pixels of region r falling in bin b , we employ

$$\sigma_{r,b} = \left[\frac{1}{|\mathcal{B}_{r,b}|} \sum_{(\mathbf{x}, t) \in \mathcal{B}_{r,b}} \Pi^*(\mathbf{x}, t)^2 \right]^{1/2} \quad (15)$$

$$\sigma_{\text{loc}}(\mathbf{x}) = \text{lerp}(\xi(\mathbf{x}); \sigma_{r(\mathbf{x}), \cdot}),$$

where lerp linearly interpolates between adjacent bin centers to avoid jumps. Dividing by σ_{loc} equalizes the target variance across the formation, near wake, and downstream wake regions. The profile is fitted on the training split only and reused unchanged at inference, so the map from $\hat{y}$ to $\hat{\Pi}$ is fixed.

B. CNN-SVGP Closure Model

The deterministic component is a compact local FiLM-CNN, following related 2D turbulence closures [12]. The CNN first predicts the standardized closure forcing y_{CNN} . An SVGP head is then trained on the CNN residual $r = y - y_{\text{CNN}}$. Its posterior mean provides an additive correction r_{SVGP} , while its predictive variance has two components: an epistemic part that grows as the local flow statistics shift from the training distribution, and a heteroscedastic noise floor accounting for

the tailedness of the residual. The corrected standardized prediction is dimensionalized to recover the closure forcing,

$$\begin{aligned} r(\mathbf{x}, t) &= y(\mathbf{x}, t) - y_{\text{CNN}}(\mathbf{x}, t), \\ \hat{y}(\mathbf{x}, t) &= y_{\text{CNN}}(\mathbf{x}, t) + r_{\text{SVGP}}(\mathbf{x}, t), \\ \hat{\Pi}(\mathbf{x}, t) &= \hat{y}(\mathbf{x}, t) \sigma_{\text{loc}}(\mathbf{x}) \frac{U(t)^2}{D_\chi^2}. \end{aligned} \quad (16)$$

The CNN-SVGP architecture and training configuration are summarized in Table II.

component	specification
<i>deterministic component</i>	
conditioning	FiLM on $(\zeta, \dot{\zeta})$; hidden width 16
CNN	[64, 64, 32], 5×5 kernels; 1×1 head; 170,481 parameters
receptive field	13×13 LES cells
optimization	standardized MSE; Adam; learning rate 10^{-3} ; weight decay 10^{-5} ; 2400 epochs
sampling	4096/1024 train/validation crops per epoch; 32×32 crops; batch size 8
<i>probabilistic residual</i>	
GP inputs	13 features: 10 flow statistics of $\bar{\omega}$, $\bar{u}$, and $\bar{v}$; y_{CNN} ; sdf; and ξ
kernel	RBF on standardized inputs; constant mean; all hyper-parameters fixed from training data before optimization
variational family	full-rank Cholesky; heteroscedastic Gaussian likelihood, per-point noise from a frozen 3×24 (region, ξ -bin) table of the training target $r(\mathbf{x}, t)$ variances
objective	variational ELBO, CNN frozen; only $(\mathbf{Z}, \mathbf{m}, \mathbf{S})$ trained
optimization	Adam; learning rate 10^{-3} ; cosine warm restarts; 100 epochs; selection by validation NLL
implementation	GPyTorch [36]

Table II. CNN-SVGP architecture and training configuration.

C. Training Procedure

A single CNN-SVGP is trained on the three training geometries and their flow ensemble data. Training crops are sampled preferentially from the dynamically active wake region, and the CNN minimizes the standardized mean-squared error

$$\mathcal{L} = \frac{1}{|\mathcal{P}|} \sum_{\mathbf{x} \in \mathcal{P}} [y_{\text{CNN}}(\mathbf{x}, t) - y(\mathbf{x}, t)]^2, \quad (17)$$

where $\mathcal{P}$ are the retained training pixels.

The SVGP is fitted to the residual in Eq. (16) using the same temporal split and population $\mathcal{P}$. Each valid point is represented by $\phi(\mathbf{x}, t) \in \mathbb{R}^{13}$: four frame-level spectral statistics, six short-window statistics within the pixel's wall-distance region (mean, standard deviation, skewness, and kurtosis of $\bar{\omega}$, and standard deviations of $\bar{u}$ and $\bar{v}$), together with y_{CNN} , sdf, and the region coordinate ξ .

The sparse posterior uses $M = 1020$ inducing variables $\mathbf{u} = f(\mathbf{Z})$ in this feature space, with $q(\mathbf{u}) = \mathcal{N}(\mathbf{m}, \mathbf{S})$ parameterized by a full-rank Cholesky factor [22, 37]. Marginalization gives

$$\begin{aligned} r_{\text{SVGP}i} &= c + \mathbf{k}_i^\top \mathbf{K}_{ZZ}^{-1} \mathbf{m}, \\ s_i^2 &= k_{ii} - \mathbf{k}_i^\top \mathbf{K}_{ZZ}^{-1} (\mathbf{K}_{ZZ} - \mathbf{S}) \mathbf{K}_{ZZ}^{-1} \mathbf{k}_i, \end{aligned} \quad (18)$$

with $q(f_i) = \mathcal{N}(\mu_i, s_i^2)$ and $\mathbf{k}_i = k(\mathbf{Z}, \phi_i)$. $\sigma_{\text{SVGP}i}^2 = s_i^2 + \sigma_n^2(\phi_i)$, where σ_n^2 is a frozen 3×24 table of the target variances indexed by region and ξ coordinate.

VI. RESULTS

A. A Priori Evaluation

We compare $\hat{\Pi}$ in Eq. (16) with the reference forcing in Eq. (9) on held-out fDNS fields. Fig. 7 shows a priori results for some in-and-out of distribution ensemble members, with point-wise correlations of 0.98 for the circular island at constant $\text{Re}_D(\bar{U})$ (in-distribution, IND), 0.92 for the square prism with chirped inlet forcing (OOD geometry and OOD inlet modulation), and 0.93 for the altered-aspect-ratio coastal cape (OOD geometry). The OOD cases broaden the scatter but remain centered around the identity line, indicating increased prediction variance without a comparable systematic bias.

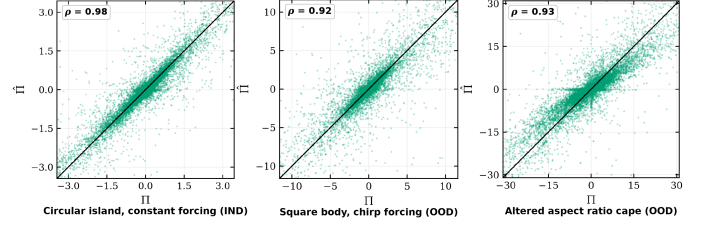


Fig. 7. A priori closure predictions for one in-distribution and two OOD configurations.

B. A posteriori Evaluation

A posteriori rollouts compare LES-CNN-SVGP with fDNS, a no-model coarse simulation (LES-NM), and a static Smagorinsky baseline (LES-SMAG) [5]. For LES-SMAG, we use $\nu_t = (C_s \bar{\Delta})^2 |S|$ with $C_s = 0.15$ following Pope [1] for every case, where $|S| = (2S_{ij}S_{ij})^{1/2}$ is the strain-rate magnitude. Nearby values of C_s were also tested but did not qualitatively improve the baseline, so C_s is fixed and not re-tuned case by case. The a posteriori LES-CNN-SVGP rollout is illustrated in Fig. 8: at each time step, the model predicts the closure forcing from the current LES state, and this forcing is explicitly incorporated into the solver's Adams–Bashforth step to advance the state.

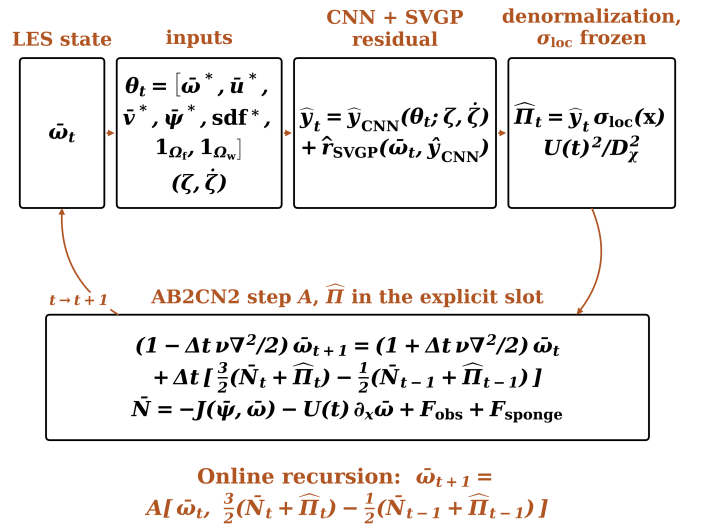


Fig. 8. A posteriori LES-CNN-SVGP rollout procedure.

Figs. 9–12 show a posteriori OOD results. LES-CNN-SVGP consistently outperforms LES-NM and LES-SMAG in reproducing the fDNS dynamics, integral quantities, and turbulent statistics across the tested distribution shifts.

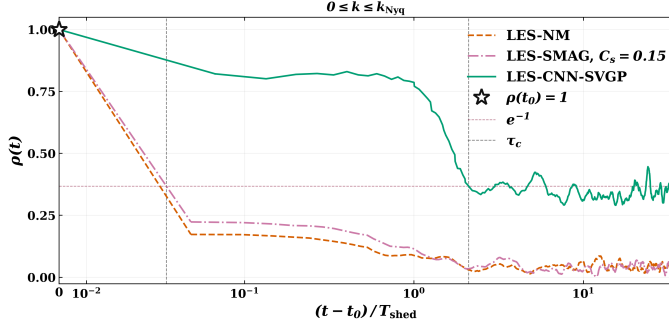


Fig. 9. A posteriori vorticity-field correlation $\rho(\omega_{\text{fDNS}}, \omega)(t)$ for the OOD submarine case, unseen $\text{Re}_L(\bar{U}) = 50600$, over a prediction horizon of approximately $30 T_{\text{shed}}$.

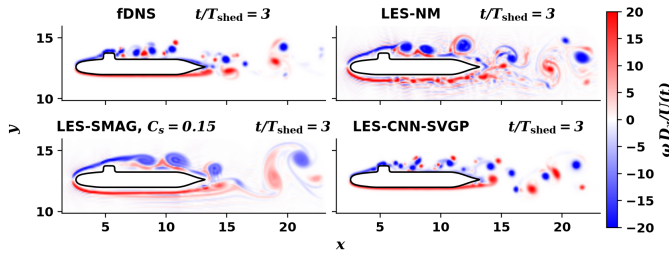


Fig. 10. A posteriori instantaneous vorticity fields at $t/T_{\text{shed}} = 3$ for the submarine case with an OOD Reynolds number, $\text{Re}_L(\bar{U}) = 50600$.

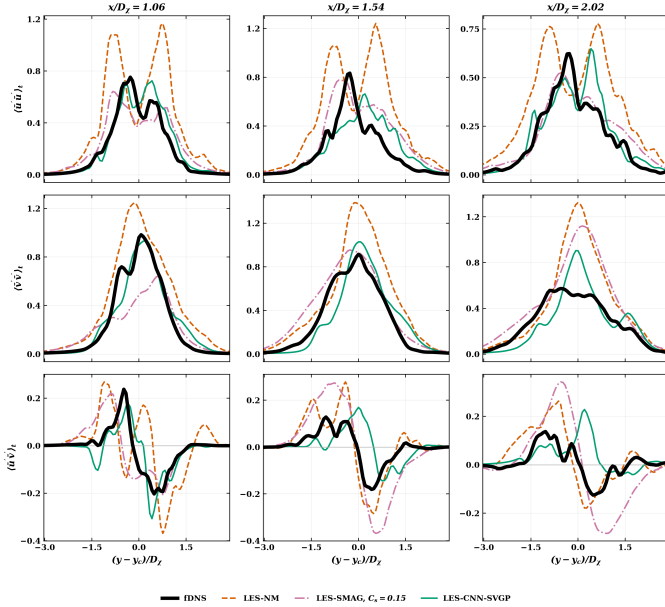


Fig. 11. Reynolds stresses flow past square prism (OOD geometry) with OOD forcing: $U(t) = \bar{U}[1 + a_U \sin(2\pi f_0 \tau(1 + \tau/T_{\text{sw}}))]$, $\tau = t - T_w$, $f_0 = 1/P_1$, $\bar{u}' = \bar{u}/U(t) - \langle \bar{u}/U(t) \rangle_t$

Fig. 13 shows that the predictive uncertainty σ_{SVGP} remains low over the ING flow but rises in OOD inference, indicating that the SVGP reliably flags out-of-distribution conditions.

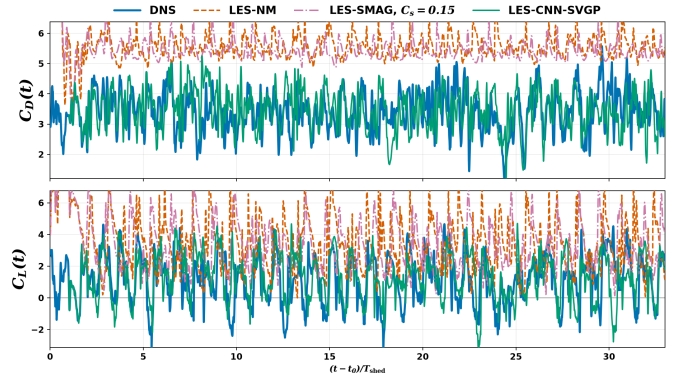


Fig. 12. A posteriori drag and lift time histories for the altered-aspect-ratio cape (OOD geometry) at $\text{Re}_{h_c}(\bar{U}) = 8500$, $T_{\text{final}} \approx 30 T_{\text{shed}}$

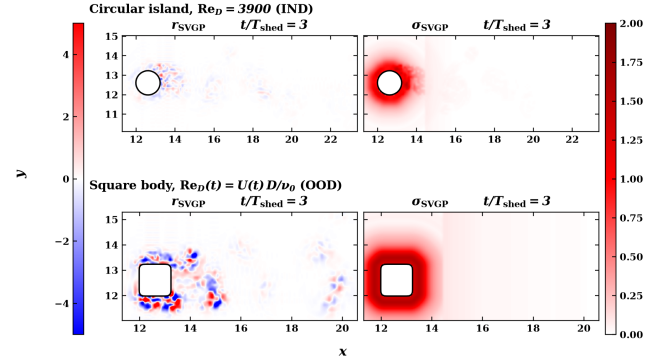


Fig. 13. SVGP residual correction r_{SVGP} and predictive standard deviation σ_{SVGP} at $t/T_{\text{shed}} = 3$, IND and OOD cases.

VII. CONCLUSIONS

We developed a FiLM-CNN-SVGP closure for nonstationary 2D flows, trained across a circular island, coastal cape, and submarine, and evaluated under shifts in geometry, Reynolds number, and background flow history. A priori evaluations showed strong point-wise agreement with the reference closure, while a posteriori LES-CNN-SVGP rollouts remained stable and better reproduced fDNS dynamics, integral quantities, and turbulent statistics than the no-model and static Smagorinsky baselines across distribution shifts. After trajectory decorrelation, the learned closure continued to recover the filtered reference statistics.

Possible future work includes neural closure models for realistic flows [38–45]. Applications with real vehicles with path planning and energy optimization [46–50] are also promising, as well as Lagrangian or bias correction applications [51–53]. Finally, GP closures allow merging with uncertainty quantification [54–56].

VIII. ACKNOWLEDGMENTS

We thank the members of our MIT-MSEAS group, especially A.N. Suresh Babu and A. Sadam for their help with the vorticity-stream-function solver and neural closure modeling. We are grateful to the Office of Naval Research for partial support under grant N00014-20-1-2023 (MURI ML-SCOPE) to the Massachusetts Institute of Technology.

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